OLA Data Analyst Project

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1. Dataset Generation Prompt
Please create a spreadsheet with 1 lac rows, for Bengaluru city. Give the following columns. The data will be for 1 month. use the following column

1. Date
2. Time
3. Booking ID
4. Booking Status
5. Customer ID
6. Vehicle Type (Auto, Prime Plus, Prime Sedan, Mini, Bike, eBike, Prime SUV)
7. Pickup Location (Dummy location points from 50 areas in Bangalore)
8. Drop Location (From dummy pickup locations)
9. Avg VTAT (Time taken to arrive at the vehicle)
10. Avg CTAT (Time taken to arrive at the Customer)
11. Cancelled Rides by Customer
12. Reason for cancelling by Customer:
    a. Driver is not moving towards pickup location
    b. Driver asked to cancel
    c. AC is not working (Only for 4-wheelers)
    d. Change of plans
    e. Wrong Address
13. Cancelled Rides by Driver
    a. Personal & Car related issues
    b. Customer related issue
    c. The customer was coughing/sick
    d. More than permitted people
14. Incomplete Rides
15. Incomplete Rides Reason:
    a. Customer Demand
    b. Vehicle Breakdown
    c. Other Issue
16. Booking Value
17. Ride Distance
18. Driver Ratings
19. Customer Rating

Data Constraints & Business Rules

- Overall successful booking rate: 62% (Fare, ratings, VTAT, CTAT only exist for successful bookings).

- Customer cancellation rate: <= 7%.
- Driver cancellation rate: <= 18%.
- Incomplete ride rate: < 6%.
- Booking ID format: 10 digits starting with 'CNR' (e.g. CNR1234567).
- Value Distribution: 70% under 500, 28% between 500-1000, 2% above 1000. Higher order values on weekends.
- Increased order volume on weekends and specified match days.

2. SQL Analysis Questions

1. Retrieve all successful bookings.
2. Find the average ride distance for each vehicle type.
3. Get the total number of cancelled rides by customers.
4. List the top 5 customers who booked the highest number of rides.
5. Get the number of rides cancelled by drivers due to personal and car-related issues.
6. Find the maximum and minimum driver ratings for Prime Sedan bookings.
7. Retrieve all rides where payment was made using UPI.
8. Find the average customer rating per vehicle type.
9. Calculate the total booking value of rides completed successfully.
10. List all incomplete rides along with the reason.

3. Power BI Dashboard Requirements

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. Cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

4. Database Schema / Data Columns

1. Date
2. Time
3. Booking_ID
4. Booking_Status
5. Customer_ID
6. Vehicle_Type

7. Pickup_Location
8. Drop_Location
9. V_TAT
10. C_TAT
11. cancelled_Rides_by_Customer
12. cancelled_Rides_by_Driver
13. Incomplete_Rides
14. Incomplete_Rides_Reason
15. Booking_Value
16. Payment_Method
17. Ride_Distance
18. Driver_Ratings
19. Customer_Rating

5. SQL Views & Solution Queries

```SQL
CREATE DATABASE Ola;
USE Ola;

-- #1. Retrieve all successful bookings:
CREATE VIEW Successful_Bookings AS
SELECT * FROM bookings
WHERE Booking_Status = 'Success';

-- #2. Find the average ride distance for each vehicle type:
CREATE VIEW ride_distance_for_each_vehicle AS
SELECT Vehicle_Type, AVG(Ride_Distance) AS avg_distance
FROM bookings
GROUP BY Vehicle_Type;

-- #3. Get the total number of cancelled rides by customers:
CREATE VIEW cancelled_rides_by_customers AS
SELECT COUNT(*) FROM bookings
WHERE Booking_Status = 'cancelled by Customer';

-- #4. List the top 5 customers who booked the highest number of
rides:
CREATE VIEW Top_5_Customers AS
```

```sql
SELECT Customer_ID, COUNT(Booking_ID) AS total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC LIMIT 5;

-- #5. Get the number of rides cancelled by drivers due to personal
and car-related issues:
CREATE VIEW Rides_cancelled_by_Drivers_P_C_Issues AS
SELECT COUNT(*) FROM bookings
WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';

-- #6. Find the maximum and minimum driver ratings for Prime Sedan
bookings:
CREATE VIEW Max_Min_Driver_Rating AS
SELECT MAX(Driver_Ratings) AS max_rating, MIN(Driver_Ratings) AS
min_rating
FROM bookings
WHERE Vehicle_Type = 'Prime Sedan';

-- #7. Retrieve all rides where payment was made using UPI:
CREATE VIEW UPI_Payment AS
SELECT * FROM bookings
WHERE Payment_Method = 'UPI';

-- #8. Find the average customer rating per vehicle type:
CREATE VIEW AVG_Cust_Rating AS
SELECT Vehicle_Type, AVG(Customer_Rating) AS avg_customer_rating
FROM bookings
GROUP BY Vehicle_Type;

-- #9. Calculate the total booking value of rides completed
successfully:
CREATE VIEW total_successful_ride_value AS
SELECT SUM(Booking_Value) AS total_successful_ride_value
FROM bookings
WHERE Booking_Status = 'Success';
```

```sql
-- #10. List all incomplete rides along with the reason:
CREATE VIEW Incomplete_Rides_Reason AS
SELECT Booking_ID, Incomplete_Rides_Reason
FROM bookings
WHERE Incomplete_Rides = 'Yes';
```

6. Power BI Dashboard Breakdown

1. Overall View

- Ride Volume Over Time: A time-series chart showing the number of rides per day/week.
- Booking Status Breakdown: A pie or doughnut chart displaying the proportion of different booking statuses (success, cancelled by customer, cancelled by driver, etc.).

2. Vehicle Type View

- Top 5 Vehicle Types by Ride Distance: A bar chart ranking vehicle types based on the total distance covered.

3. Revenue View

- Revenue by Payment Method: A stacked bar chart displaying total revenue based on payment methods (Cash, UPI, Credit Card, etc.).
- Top 5 Customers by Total Booking Value: A leaderboard visual listing customers who have spent the most on bookings.
- Ride Distance Distribution Per Day: A histogram or scatter plot showing the distribution of ride distances for different dates.

4. Cancellation View

- Cancelled Rides Reasons: A bar chart highlighting the common reasons for ride cancellations by customers and drivers.

5. Ratings View

- Average Customer Ratings by Vehicle Type: A column chart showing the average customer ratings for different vehicle types.

- Driver Ratings Distribution: A box plot visualizing the spread of driver ratings for different vehicle types.

Customer vs. Driver Ratings: A scatter plot comparing customer and driver ratings for each completed ride to analyze correlations.